Hybrid experiment theoretical essentials

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Contents

1 Standing wave optical cavity in vacuum

1.1 Reflection and transmission at an interface

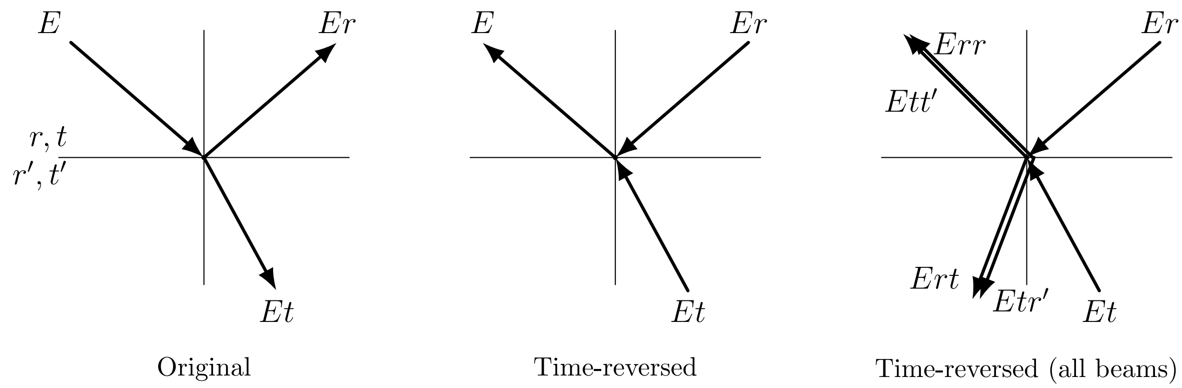

Figure 1: Original and time-reversed interaction between a beam and an interface

The first question we need to answer before moving on to calculating reflected, transmitted and intracavity fields, is the following. If light is shined to an interface from one side and the interface has reflected coefficient r and transmitted coefficient t, what would be the reflected and transmitted coefficients r' and t' if the light is shined from another side? This elegant derivation [1] of the relation between the field transmission and reflection coefficients was made by G. G. Stokes. He assumed that there was no dissipation at the interface and therefore the process obeyed time-reversal invariance. The three waves are shown on the left in the figure 1. The time-reversed case appears in the center plot. When we reverse time, we need to include three additional waves: the reflection from the bottom wave, and the transmission of both the bottom and top waves. All of the waves are shown in the plot on the right. In order for the right-hand plot to be equivalent to the time-reversed center plot, the two waves in the upper left must sum to E and the two waves in the lower left must sum to zero. Thus,

$$E = Ett' + Err \tag{1.1}$$
$$0 = Ert + Er't \tag{1.2}$$

from which we conclude that

$$t' = \frac{1-r^2}{t} \tag{1.3}$$
$$r' = -r \tag{1.4}$$

Here we assume that $r, r', t, t' \in \mathbb{C}$. The fact, that we assumed no dissipation should mean, that

$$|t|^2 + |r|^2 = 1 \tag{1.5}$$
$$|t'|^2 + |r'|^2 = 1 \tag{1.6}$$

Let's plug 1.4 and 1.3 to 1.6. After some algebra we arrive to

$$\frac{1}{|t|^2}(|1-r^2|^2 - (1-|r|^2)^2) = 0 \tag{1.7}$$

Since $|r| \leq 1$, ($r = |r|e^{i\varphi_r}$) it should mean that

$$|1-r^2| = 1-|r|^2 = \sqrt{1-2|r|^2\cos(2\varphi_r)+|r|^4} \tag{1.8}$$

Equation 1.8 is satisfied $\forall\ |r| \leq 1$ only if $\varphi_r = 0$ or π. That means that $r, r' \in \mathbb{R}$. Then ($t = |t|e^{i\varphi_t}$)

$$t' = \frac{|t|^2}{t} = |t|e^{-i\varphi_t} = t^* \tag{1.9}$$

So finally we have

$$r' = -r, \quad r, r' \in \mathbb{R} \tag{1.10}$$

$$t' = t^*, \quad t, t' \in \mathbb{C} \tag{1.11}$$

1.2 Reflected, transmitted and intracavity fields and intensities

Let us start calculating the reflected, transmitted and intracavity fields. Figure 2 depicts the schematic of the electromagnetic wave traveling inside the optical cavity. We assume that the mirrors 1 and 2 have corresponding amplitude transmission and reflection coefficients r_1, t_1 and r_2, t_2. The intracavity medium is lossless, however there is a loss of amplitude due to scattering at the mirror surfaces: t_a at Mirror 1 and t_b at Mirror 2 ($t = t_a t_b$). This model is the closest approximation of a two-mirror cavity inside ultra-high vacuum environment.

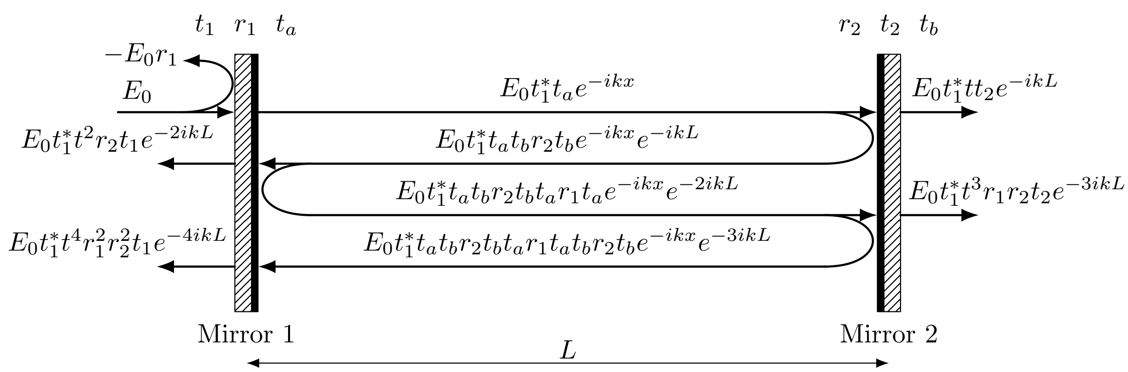

Figure 2: Optical cavity schematic

1.2.1 Reflected field and intensity

If the incident amplitude of the field is E_0, by summing all the paths that contribute to the reflected field and using 1.10, 1.11 we arrive to the following series (we relabel $t = t_a t_b$)

$$\begin{aligned} E_r &= E_0(-r_1 + t_1^* t r_2 t t_1 e^{-2ikL} + t_1^* t r_2 t r_1 t r_2 t t_1 e^{-4ikL} + ...) = \\ &= E_0(-r_1 + |t_1|^2 t^2 r_2 e^{-2ikL}(1 + t^2 r_1 r_2 e^{-2ikL} + (t^2 r_1 r_2 e^{-2ikL})^2 + ...)) = \\ &= E_0(-r_1 + \frac{|t_1|^2 t^2 r_2 e^{-2ikL}}{1 - t^2 r_1 r_2 e^{-2ikL}}) = E_0 \frac{t^2 r_2 e^{-2ikL} - r_1}{1 - t^2 r_1 r_2 e^{-2ikL}} \end{aligned}$$

We obtain an expression for the reflected field:

$$E_r = E_0 \frac{t^2 r_2 e^{-2ikL} - r_1}{1 - t^2 r_1 r_2 e^{-2ikL}} \tag{1.12}$$

Let's remember that

$$FSR = \frac{c}{2L} \tag{1.13}$$

$$kL = \frac{2\pi L f}{c} = \frac{\pi f}{FSR} = \pi \tilde{f} \tag{1.14}$$

where $\tilde{f} = \frac{f}{FSR}$ is a normalized frequency. After some algebra we get an expression for reflected intensity:

$$I_r = I_0 \frac{(r_1 - t^2 r_2)^2 + 4t^2 r_1 r_2 \sin^2(\pi \tilde{f})}{(1 - t^2 r_1 r_2)^2 + 4t^2 r_1 r_2 \sin^2(\pi \tilde{f})} \tag{1.15}$$

The minimum reflected intensity is obtained on resonance (when $\tilde{f} \in \mathbb{N}$):

$$I_r^{res} = I_0 \frac{(r_1 - t^2 r_2)^2}{(1 - t^2 r_1 r_2)^2} \tag{1.16}$$

1.2.2 Transmitted field and intensity

If the incident amplitude of the field is E_0, by summing all the paths that contribute to the transmitted field and using 1.10, 1.11 we arrive to the following series (we relabel $t = t_a t_b$)

$$\begin{aligned} E_t &= E_0(t_1^* t t_2 e^{-ikL} + t_1^* t r_2 t r_1 t t_2 e^{-3ikL} + ...) = \\ &= E_0 t_1^* t t_2 e^{-ikL}(1 + t^2 r_1 r_2 e^{-2ikL} + (t^2 r_1 r_2 e^{-2ikL})^2 + ...) = \\ &= E_0 \frac{t_1^* t t_2 e^{-ikL}}{1 - t^2 r_1 r_2 e^{-2ikL}} \end{aligned}$$

We obtain an expression for the transmitted field:

$$E_t = E_0 \frac{t_1^* t t_2 e^{-ikL}}{1 - t^2 r_1 r_2 e^{-2ikL}} \tag{1.17}$$

After some algebra we get an expression for transmitted intensity ($\tilde{f} = f/FSR$):

$$I_t = I_0 \frac{|t_1|^2 |t_2|^2 t^2}{(1 - t^2 r_1 r_2)^2 + 4t^2 r_1 r_2 \sin^2(\pi \tilde{f})} \tag{1.18}$$

The maximum transmitted intensity is obtained on resonance (when $\tilde{f} \in \mathbb{N}$):

$$I_t^{res} = I_0 \frac{|t_1|^2 |t_2|^2 t^2}{(1 - t^2 r_1 r_2)^2} \tag{1.19}$$

1.2.3 Intracavity field and intensity

We need to calculate fields propagating in the direction of incident light ($E_c^+(x)$), in the opposite direction ($E_c^-(x)$) and full intracavity field ($E_c(x)$). The process of calculation is very similar to the reflected and transmitted fields calculations, but this time we have to take care of at which point in the cavity the field is being calculated. Let's pick a point at a distance x from the first mirror and calculate forward and backward fields there. For the forward field:

$$\begin{aligned} E_c^+(x) &= E_0(t_1^* t_a e^{-ikx} + t_1^* t_a t_b r_2 t_b t_a r_1 t_a e^{-ikx} e^{-2ikL} + ...) = \\ &= E_0 t_1^* t_a e^{-ikx}(1 + t^2 r_1 r_2 e^{-2ikL} + (t^2 r_1 r_2 e^{-2ikL})^2 + ...) = \\ &= E_0 \frac{t_1^* t_a e^{-ikx}}{1 - t^2 r_1 r_2 e^{-2ikL}} \end{aligned}$$

We obtain:

$$E_c^+(x) = E_0 \frac{t_1^* t_a e^{-ikx}}{1 - t^2 r_1 r_2 e^{-2ikL}} \tag{1.20}$$

The forward propagating intensity is:

$$I_c^+ = I_0 \frac{|t_1|^2 t_a^2}{(1 - t^2 r_1 r_2)^2 + 4t^2 r_1 r_2 \sin^2(\pi \tilde{f})} \tag{1.21}$$

For the backward field:

$$\begin{aligned} E_c^-(x) &= E_0(t_1^* t_a t_b r_2 t_b e^{-ik(2L-x)} + t_1^* t_a t_b r_2 t_b t_a r_1 t_a t_b r_2 t_b e^{-ik(4L-x)} + ...) = \\ &= E_0 t_1^* t r_2 t_b e^{-ik(2L-x)}(1 + t^2 r_1 r_2 e^{-2ikL} + (t^2 r_1 r_2 e^{-2ikL})^2 + ...) = \\ &= E_0 \frac{t_1^* t r_2 t_b e^{-ik(2L-x)}}{1 - t^2 r_1 r_2 e^{-2ikL}} \end{aligned}$$

It's here that we have to acknowledge that the amplitude reflection coefficients are actually negative, because upon reflection from a medium with higher refractive index the light undergoes a π phase shift. So if we take this into account we obtain:

$$E_c^-(x) = E_0 \frac{-t_1^* t |r_2| t_b e^{-ik(2L-x)}}{1 - t^2 r_1 r_2 e^{-2ikL}} \tag{1.22}$$

Full intracavity field:

$$E_c(x) = E_c^+(x) + E_c^-(x) = E_0 \frac{t_1^* t_a e^{-ikx} - t_1^* t |r_2| t_b e^{-ik(2L-x)}}{1 - t^2 r_1 r_2 e^{-2ikL}}$$

We obtain:

$$E_c(x) = E_0 \frac{t_1^* (t_a e^{ik(L-x)} - t t_b |r_2| e^{-ik(L-x)}) e^{-ikL}}{1 - t^2 r_1 r_2 e^{-2ikL}} \tag{1.23}$$

After some algebra we get an expression for intracavity intensity ($\tilde{f} = f/FSR$, $\tilde{x} = 2x/\lambda$):

$$I_c(\tilde{x}) = I_0 \frac{|t_1|^2((t_a + t t_b |r_2|)^2 - 4t^2 |r_2| \cos^2(\pi(\tilde{x} - \tilde{f})))}{(1 - t^2 r_1 r_2)^2 + 4t^2 r_1 r_2 \sin^2(\pi \tilde{f})} \tag{1.24}$$

The maximum internal intensity is obtained on resonance (when $\tilde{x} = m + 1/2, \quad \tilde{f}, m \in \mathbb{N}$):

$$I_c^{res,max} = I_0 \frac{|t_1|^2 (t_a + t t_b |r_2|)^2}{(1 - t^2 r_1 r_2)^2} \approx I_0 \frac{4|t_1|^2}{(1 - t^2 r_1 r_2)^2} \tag{1.25}$$

The propagating internal intensity on resonance:

$$I_c^{res,+} = I_0 \frac{|t_1|^2 t_a^2}{(1 - t^2 r_1 r_2)^2} \approx I_0 \frac{|t_1|^2}{(1 - t^2 r_1 r_2)^2} \tag{1.26}$$

1.3 Measuring cavity parameters

Parameter of the cavity, characterizing the "goodness" of the cavity is called *finesse* ($\mathcal{F}$). It's defined as follows:

$$\mathcal{F} := \frac{FSR}{\gamma} \tag{1.27}$$

where γ is full-width at half-maximum (FWHM) of transmitted or internal intensities. If we now equate

$$I_t(\tilde{f}_{1/2}) = \frac{I_t^{res}}{2} \tag{1.28}$$

After some algebra and series expansions (since $1 - t^2 r_1 r_2 \ll 1$) we get:

$$\tilde{f}_{1/2} = \frac{1 - t^2 r_1 r_2}{2\pi} = \frac{\gamma}{2FSR} \tag{1.29}$$

$$\mathcal{F} = \frac{\pi}{1 - t^2 r_1 r_2} \tag{1.30}$$

Finesse is also related to the intensity enhancement inside of the cavity:

$$\frac{I_c^{res}}{I_0} = 4 \frac{1 - r_1^2}{(1 - t^2 r_1 r_2)^2} \tag{1.31}$$

In a lossless and symmetric ($r_1 = r_2, \ t = 1$) case it simply yields:

$$\frac{I_c^{res}}{I_0} = \frac{4\mathcal{F}}{\pi} \tag{1.32}$$

Please take into account that "lossless and symmetric ($r_1 = r_2,\ t = 1$) case" is never actually achievable in real life, and the results 1.32 for ideal case and 1.31 for a real case vary **significantly**, sometimes **by several orders of magnitude**.

If one represents $|t|, |r_1|, |r_2|$ as follows:

$$|r_1| = 1 - a,\ a \ll 1 \tag{1.33}$$
$$|r_2| = 1 - b,\ b \ll 1 \tag{1.34}$$
$$|t| = 1 - c,\ c \ll 1 \tag{1.35}$$

it becomes possible to make series expansions of all the quantities derived above. After some algebra it turns out that if one experimentally measures finesse, reflected intensity on resonance and transmitted intensity on resonance, intracavity intensity, mirror parameters and cavity losses can be derived from these three parameters with simple expressions:

$$\frac{I_c^{res}}{I_0} = \frac{4\mathcal{F}}{\pi}(1 - \sqrt{R_c}) \tag{1.36}$$
$$\frac{I_c^{res,+}}{I_0} = \frac{\mathcal{F}}{\pi}(1 - \sqrt{R_c}) \tag{1.37}$$
$$T_1 = \pi\frac{1 - \sqrt{R_c}}{\mathcal{F}} \tag{1.38}$$
$$T_2 = \pi\frac{T_c}{\mathcal{F}(1 - \sqrt{R_c})} \tag{1.39}$$
$$A_L = 1 - (T_a T_b)^2 = \pi\frac{1 - R_c - T_c}{\mathcal{F}(1 - \sqrt{R_c})} \tag{1.40}$$

A_L here is a round-trip loss coefficient (without mirror leakage). If we want to calculate the overall steady-state loss coefficient its given by a trivial expression (which is consistent with the equations above):

$$A_c = A_L\frac{I_c^{res,+}}{I_0} = 1 - R_c - T_c \tag{1.41}$$

We can also express intracavity power, finesse and cavity reflection and transmission through mirror parameters and cavity loss:

$$\frac{I_c^{res}}{I_0} = \frac{16T_1}{(A_L + T_1 + T_2)^2} \tag{1.42}$$
$$\frac{I_c^{res,+}}{I_0} = \frac{4T_1}{(A_L + T_1 + T_2)^2} \tag{1.43}$$
$$\mathcal{F} = \frac{2\pi}{A_L + T_1 + T_2} \tag{1.44}$$
$$R_c = \left(\frac{A_L - T_1 + T_2}{A_L + T_1 + T_2}\right)^2 \tag{1.45}$$
$$T_c = \frac{4T_1T_2}{(A_L + T_1 + T_2)^2} \tag{1.46}$$

1.4 Mode matching

Usually, depending on the alignment of the input beam with respect to the cavity axis, the beam waist and wavefront matching, and the quality of the optical components, only a fraction of the incident light couples into the desired cavity mode (usually the fundamental TEM_{00} mode). This fraction is referred to as the mode-matching efficiency:

$$\eta = \frac{I_{00}}{I_0},\ I_0 = I_{00} + I_h \tag{1.47}$$

The remaining intensity, I_h, consists of two contributions. One part couples into higher-order Hermite-Gaussian cavity modes, while the other does not couple to any cavity mode and is directly reflected from the input mirror. An ideal paraxial beam can always be expanded in the HG basis of the cavity. In practice, however, imperfections such as non-paraxial field components, clipping, aberrations reduce the overlap with the cavity eigenmodes, producing light that can not be decomposed into HG modes.

Taking into account the coupling efficiency η we can express apparent reflectance and transmittance of the cavity through the actual ones ($R_c = I_r^{cav}/I_{00},\ R_c^{app} = I_r^{full}/I_0,\ T_c = I_t^{cav}/I_{00},\ T_c^{app} = I_t^{full}/I_0$):

$$R_c^{app} = \eta R_c + 1 - \eta \tag{1.48}$$

$$T_c^{app} = \eta T_c \tag{1.49}$$

And reverse:

$$R_c = \frac{1}{\eta}(R_c^{app} - 1 + \eta) \tag{1.50}$$

$$T_c = \frac{1}{\eta} T_c^{app} \tag{1.51}$$

2 Cold atoms and atom-light interaction

2.1 Energy levels

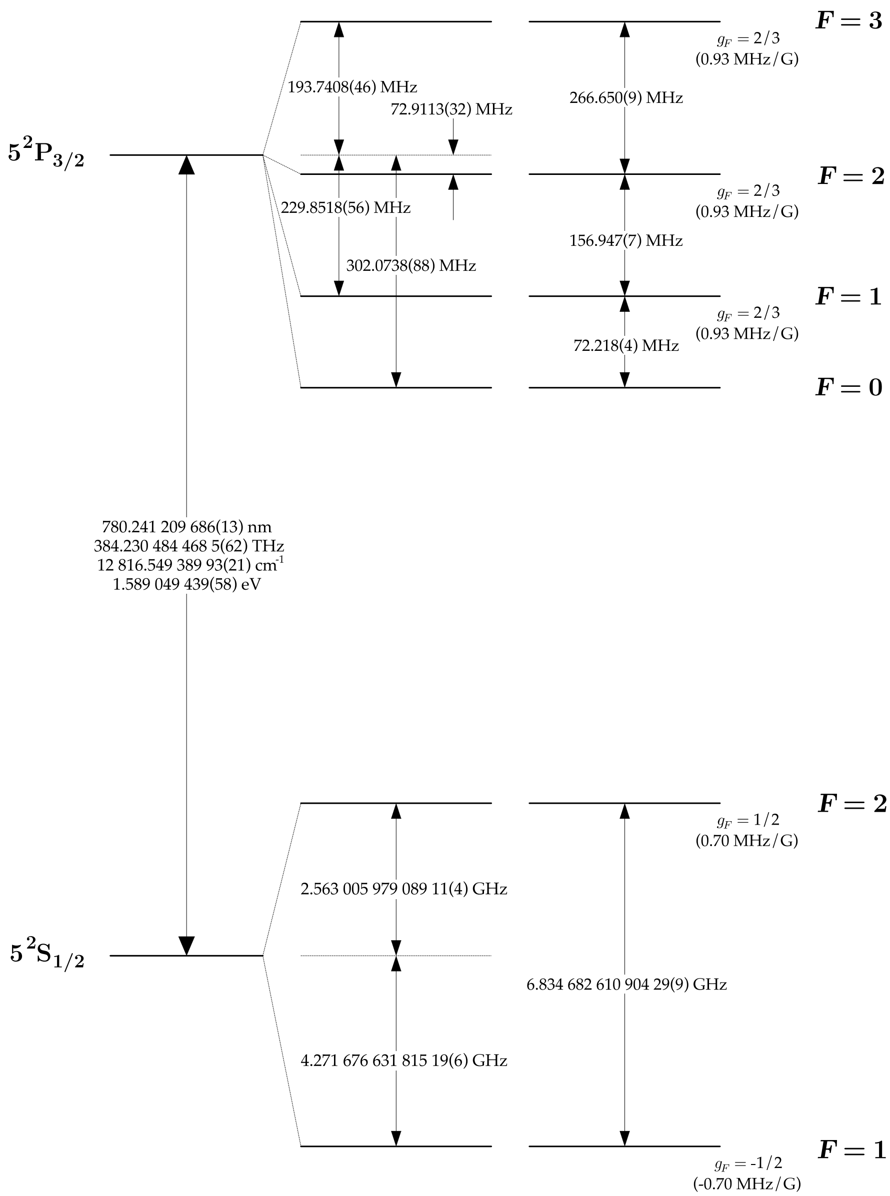

Figure 3: ^{87}Rb level diagram [2]

The experiment that will be described here will work with ^{87}Rb atoms. Specifically with the D2 line of ^{87}Rb (Fig. 3). We are mostly interested in addressing the hyperfine ground states $|g_\uparrow\rangle = |5^2S_{1/2}, F = 2, m_F = 0\rangle = |\uparrow\rangle$ and $|g_\downarrow\rangle = |5^2S_{1/2}, F = 1, m_F = 0\rangle = |\downarrow\rangle$ (so-called Rb clock states). The selection rules for electric dipole (E1) transitions say that if $\Delta L = 0$ the transition is forbidden [3]. The $|\uparrow\rangle$ and $|\downarrow\rangle$ states have the same angular momentum number $L = 0$, hence an E1 transition between them is forbidden. That's the reason why the $|\uparrow\rangle$ state has such a small decay rate ($\Gamma_\uparrow \approx 1$ mHz). It is not forbidden for the magnetic dipole (M1) transitions to happen between these two levels, meaning that it is possible to drive this transition with a microwave antenna. Since there is a manifold of excited states

$5^2P_{3/2}$ separated by an optical frequency ($\lambda = 780$ nm) from the ground $5^2S_{1/2}$ states, the system can be driven into these excited states via a laser. From now on the hyperfine levels of $5^2S_{1/2}$ will be referred to as $F = 1, 2$ and hyperfine levels of $5^2P_{3/2}$ will be referred to as $F' = 0, 1, 2, 3$. Each of the F and F' levels have a Zeeman manifold which splits the level into $2F - 1$ states. Let's for now only consider magnetically insensitive states with $m_F = 0$, implying that we are only considering transitions driven by π-polarized light. There is a selection rule [3] that says that for E1 transitions with $\Delta F = 0$, $m_F = 0 \not\leftrightarrow 0$, which means that transitions $|F = 1, m_F = 0\rangle \not\leftrightarrow |F' = 1, m_F = 0\rangle$ and $|F = 2, m_F = 0\rangle \not\leftrightarrow |F' = 2, m_F = 0\rangle$ are forbidden (effective level diagram as per Fig. 4). This means that if an atom is initially in $|\uparrow\rangle$ state it can not get to $|\downarrow\rangle$. There is no real population transfer between $\uparrow$ and $\downarrow$ subsystems, but if the system is prepared in equal population of $|\uparrow\rangle$ and $|\downarrow\rangle$, the cavity interaction can put atoms into the respective excited states and introduce apparent population transfer between $|\uparrow\rangle$ and $|\downarrow\rangle$.

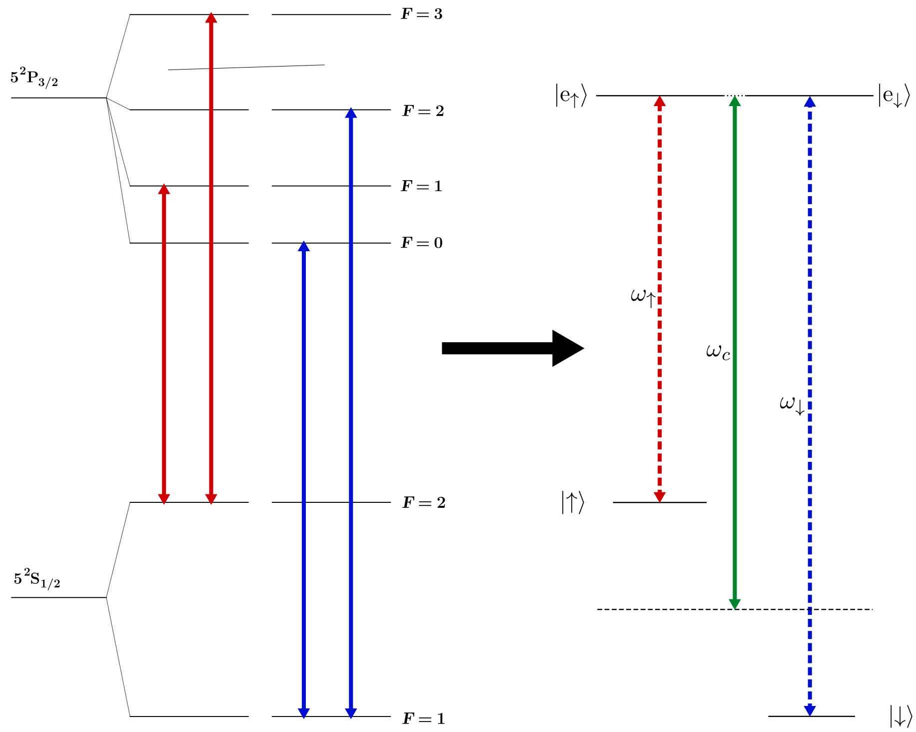

Figure 4: Effective level diagram

2.2 Atoms cooling and trapping

2.2.1 Magneto-optical trap

2.2.2 Optical molasses

2.3 Atom-light interaction

The most physically accurate way do describe atom-light interaction (see Section 2.1) for a ^{87}Rb ensemble driven with π-polarized light is to treat a Rubidium atom as two separate two-level systems interacting with a cavity mode (Fig. 5). In case of D2 line of ^{87}Rb $|g_A\rangle = |5^2S_{1/2}, F = 2, m_F = 0\rangle = |\uparrow\rangle$, $|g_B\rangle = |5^2S_{1/2}, F = 1, m_F = 0\rangle = |\downarrow\rangle$, $|e_A\rangle = |5^2S_{3/2}, F = \{1, 3\}, m_F = 0\rangle$, $|e_B\rangle = |5^2S_{3/2}, F = \{0, 2\}, m_F = 0\rangle$ [2].

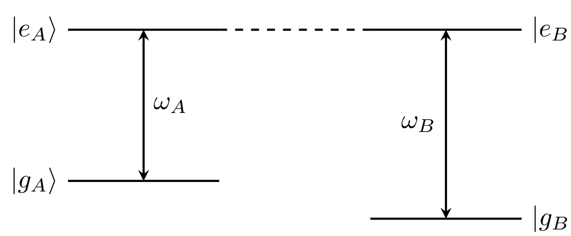

Figure 5: Level diagram

The Hamiltonian of the system with many atoms can be broken down into a non-interacting part H_0 and an interacting part H_{int}.

$$H_0 = \omega_c a^\dagger a + \sum_{i\in A} \omega_A \sigma_z^{A(i)} + \sum_{i\in B} \omega_B \sigma_z^{B(i)} \tag{2.1}$$

$$H_{\text{int}} = -\sum_{i\in A,B} \hat{d}^{(i)} \cdot \hat{E} \tag{2.2}$$

where ω_c - cavity mode frequency, a - cavity annihilation operator, $\hat{d}$ - dipole operator, $\hat{E}$ - electric field operator, $\sigma_z = |e\rangle\langle e| - |g\rangle\langle g|$, $\sigma_+ = |e\rangle\langle g|$, $\sigma_- = |g\rangle\langle e|$.

Dipole operator for a single atom can be written as

$$\hat{d} = \sum_{i,j} D_{ij} |i\rangle\langle j| \qquad (i, j \in g_A, g_B, e_A, e_B) \tag{2.3}$$

We assume the only E1 allowed transitions are $|g_A\rangle \leftrightarrow |e_A\rangle$ and $|g_B\rangle \leftrightarrow |e_B\rangle$. Then the expression for the single atom dipole operator simplifies to

$$\hat{d} = D_{ge}^A \sigma_-^A + D_{eg}^A \sigma_+^A + D_{ge}^A \sigma_-^B + D_{eg}^B \sigma_+^B \tag{2.4}$$

Without loss of generality we can set $\hat{E} = \mathcal{E}_0(a + a^\dagger)$. Since dipole operator must be Hermitian $D_{eg} = D_{ge}^*$. It turns out that its safe to assume that $D_{eg} \in \mathbb{R}$. Then we obtain expression for the interacting part of the Hamiltonian with many atoms in the following form

$$H_{\text{int}} = \sum_{i\in A} g_A (a + a^\dagger)(\sigma_+^{A(i)} + \sigma_-^{A(i)}) + \sum_{i\in B} g_B (a + a^\dagger)(\sigma_+^{B(i)} + \sigma_-^{B(i)}) \tag{2.5}$$

where $g_A = \mathcal{E}_0 \cdot D_A$, $g_B = \mathcal{E}_0 \cdot D_B$.

We can rewrite the Hamiltonian in terms of the collective spin operators

$$J_{z,+,-}^A = \sum_{i\in A} \sigma_{z,+,-}^{A(i)} \qquad \text{and} \qquad J_{z,+,-}^B = \sum_{i\in B} \sigma_{z,+,-}^{B(i)} \tag{2.6}$$

$$H_0 = \omega_c a^\dagger a + \omega_A J_z^A + \omega_B J_z^B \tag{2.7}$$

$$H_{\text{int}} = g_A (a + a^\dagger)(J_+^A + J_-^A) + g_B (a + a^\dagger)(J_+^B + J_-^B) \tag{2.8}$$

Now if we go to the interaction picture with a unitary transformation $U = \exp(itH_0)$, $H_I = U H_{\text{int}} U^\dagger$, we will see that terms of the new Hamiltonian with aJ_-^A, aJ_-^B and $a^\dagger J_+^A, a^\dagger J_+^B$ will contain fast oscillating exponents with frequencies $2\omega_c$. We apply the *rotating wave approximation* (RWA) by dropping these terms from the Hamiltonian, and then going back to the Schrödinger picture.

$$H_0 = \omega_c a^\dagger a + \omega_A J_z^A + \omega_B J_z^B \tag{2.9}$$

$$H_{\text{RWA}} = g_A(aJ_+^A + a^\dagger J_-^A) + g_B(aJ_+^B + a^\dagger J_-^B) \tag{2.10}$$

2.3.1 Effective Hamiltonian

In order to arrive at the dispersive regime Hamiltonian, we can do a following unitary transform:

$$U = e^S, \qquad S(t) = const, \qquad S^\dagger = -S \tag{2.11}$$

Then the transformed Hamiltonian:

$$H' = UHU^\dagger + i\dot{U}U^\dagger = UHU^\dagger = e^S He^{S^\dagger} = e^S He^{-S} \tag{2.12}$$

Using Baker-Campbell-Hausdorff formula we obtain

$$H' = H_0 + H_{\text{int}} + [S, H_0] + [S, H_{\text{int}}] + \frac{1}{2}[S,[S,H_0]] + \left\{ \frac{1}{2}[S,[S,H_{\text{int}}]] + \sum_{k=3}^{+\infty} \frac{[S,H]_k}{k!} \right\} \tag{2.13}$$

Let's choose S such that

$$[S, H_0] = -H_{\text{int}} \tag{2.14}$$

then

$$\begin{aligned} H' &= H_0 + H_{\text{int}} + [S, H_0] + [S, H_{\text{int}}] + \frac{1}{2}[S,[S,H_0]] + \left\{ \frac{1}{2}[S,[S,H_{\text{int}}]] + \sum_{k=3}^{+\infty} \frac{[S,H]_k}{k!} \right\} = \\ &= H_0 + \frac{1}{2}[S, H_{\text{int}}] + \left\{ \frac{1}{2}[S,[S,H_{\text{int}}]] + \sum_{k=3}^{+\infty} \frac{[S,H]_k}{k!} \right\} \end{aligned} \tag{2.15}$$

This expression might look somewhat intimidating, since it has an infinite number of commutators which we need to compute to obtain the final result. But let's not lose hope just yet and compute the first commutator $[S, H_{\text{int}}]$. One can show that S that satisfies these conditions can be written in a form

$$S = \lambda_A g_A(aJ_+^A - a^\dagger J_-^A) + \lambda_B g_B(aJ_+^B - a^\dagger J_-^B) \tag{2.16}$$

where λ_A and λ_B can be found from substituting 2.16 into 2.14. After some algebra we obtain:

$$\lambda_{A,B} = -\frac{1}{\Delta_{A,B}}, \qquad \Delta_{A,B} = \omega_c - \omega_{A,B} \tag{2.17}$$

$$S = -\frac{g_A}{\Delta_A}(aJ_+^A - a^\dagger J_-^A) - \frac{g_B}{\Delta_B}(aJ_+^B - a^\dagger J_-^B) \tag{2.18}$$

From the expression for S (2.18), we can see that actually the rest of the series in curly brackets will be $o(g_A/\Delta_A, g_B/\Delta_B)$. This fact gives us the condition for the convergence of this series and for neglecting the higher order terms (dispersive regime):

$$H_{\text{disp}} = H_0 + \frac{1}{2}[S, H_{\text{int}}], \quad \text{if} \quad |\Delta_A| \gg g_A, |\Delta_B| \gg g_B \tag{2.19}$$

Now the final step is to compute $[S, H_{\text{int}}]$. After some algebra we obtain the expression

$$[S, H_{\text{int}}] = -\sum_{\alpha=A,B} \frac{2g_\alpha^2}{\Delta_\alpha}(a^\dagger a J_z^\alpha + J_+^\alpha J_-^\alpha) - g_A g_B \left(\frac{1}{\Delta_A} + \frac{1}{\Delta_B}\right)(J_+^A J_-^B + J_-^A J_+^B) \tag{2.20}$$

The first part of this expression is the dispersive atom-cavity interaction Hamiltonian. The second one is cavity-mediated spin-spin interaction. This transformation is known as Schrieffer-Wolff transformation. We can write down a full Hamiltonian:

$$H_{\text{eff}} = \omega_c a^\dagger a + \omega_A J_z^A + \omega_B J_z^B - \sum_{\alpha=A,B} \frac{g_\alpha^2}{\Delta_\alpha}(a^\dagger a J_z^\alpha + J_+^\alpha J_-^\alpha) - \frac{g_A g_B}{2}\left(\frac{1}{\Delta_A} + \frac{1}{\Delta_B}\right)(J_+^A J_-^B + J_-^A J_+^B) \tag{2.21}$$

Now we can re-parametrize the variables so that we have more experimentally-important variables

$$A \to \uparrow, \qquad B \to \downarrow \tag{2.22}$$

$$\omega_A \to \omega_e - \frac{\omega_a}{2}, \qquad \omega_B \to \omega_e + \frac{\omega_a}{2} \tag{2.23}$$

$$\Delta = \omega_c - \omega_e, \qquad \Delta_A \to \Delta + \frac{\omega_a}{2}, \qquad \Delta_B \to \Delta - \frac{\omega_a}{2} \tag{2.24}$$

$$\sigma_z = |g_\uparrow\rangle\langle g_\uparrow| - |g_\downarrow\rangle\langle g_\downarrow| = \sigma_z^B - \sigma_z^A + \sigma_+^A\sigma_-^A - \sigma_+^B\sigma_+^B \tag{2.25}$$

$$J_z = \frac{1}{2}(J_z^B - J_z^A + J_+^A J_-^A - J_+^B J_-^B) \tag{2.26}$$

$$\sigma_N = |g_\uparrow\rangle\langle g_\uparrow| + |g_\downarrow\rangle\langle g_\downarrow| = -\sigma_z^B - \sigma_z^A + \sigma_+^A\sigma_-^A + \sigma_+^B\sigma_+^B \tag{2.27}$$

$$J_N = \frac{1}{2}(-J_z^B - J_z^A + J_+^A J_-^A + J_+^B J_-^B) \tag{2.28}$$

Terms like $J_+^A J_-^A$ are proportional to the corresponding excited state population. We will assume that we can neglect those terms because the population of both excited levels is negligible ($M_e^A, M_e^B \ll N$) because of spontaneous decay of the excited state, that has lifetime of $\tau = 26.24$ ns [2]. Furthermore, terms that contain J_N are just constants proportional to number of atoms N, so we can get rid of them in the Hamiltonian too. For π-polarized light, Clebsch-Gordan coefficients for the relevant transitions are listed in tables 1 and 2. Since the quantity entering the full Hamiltonian (2.21) is g^2, we can calculate that $g_A^2 \sim 1/30 + 3/10 = 1/3$ and $g_B^2 \sim 1/6 + 1/6 = 1/3$. That means that we can pretty safely assume that $g_A = g_B = g$.

After applying all of those assumptions we get the effective Hamiltonian:

$$H_{\text{eff}} = \omega_c a^\dagger a + \omega_a J_z + \frac{4g^2}{\omega_a(1 - \frac{4\Delta^2}{\omega_a^2})} a^\dagger a J_z \tag{2.29}$$

	$m_F = -1$	$m_F = 0$	$m_F = 1$
$F' = 2$	$-\sqrt{\frac{1}{8}}$	$-\sqrt{\frac{1}{6}}$	$-\sqrt{\frac{1}{8}}$
$F' = 1$	$-\sqrt{\frac{5}{24}}$	0	$\sqrt{\frac{5}{24}}$
$F' = 0$		$\sqrt{\frac{1}{6}}$	

Table 1: ^{87}Rb D_2 dipole matrix elements for π transitions ($F = 1, m_F \to F', m_F' = m_F$), expressed as multiples of $\langle J = \frac{1}{2}|er|J' = \frac{3}{2}\rangle$.

	$m_F = -2$	$m_F = -1$	$m_F = 0$	$m_F = 1$	$m_F = 2$
$F' = 3$	$-\sqrt{\frac{1}{6}}$	$-\sqrt{\frac{4}{15}}$	$-\sqrt{\frac{3}{10}}$	$-\sqrt{\frac{4}{15}}$	$-\sqrt{\frac{1}{6}}$
$F' = 2$	$-\sqrt{\frac{1}{6}}$	$-\sqrt{\frac{1}{24}}$	0	$\sqrt{\frac{1}{24}}$	$\sqrt{\frac{1}{6}}$
$F' = 1$		$\sqrt{\frac{1}{40}}$	$\sqrt{\frac{1}{30}}$	$\sqrt{\frac{1}{40}}$	

Table 2: ^{87}Rb D_2 dipole matrix elements for π transitions ($F = 2, m_F \to F', m'_F = m_F$), expressed as multiples of $\langle J = \frac{1}{2}|er|J' = \frac{3}{2}\rangle$.

2.3.2 One-axis twisting

3 Optomechanics

3.1 Cavity optomechanics

3.1.1 Linearized Hamiltonian

3.1.2 Optical spring

3.1.3 Sideband cooling

3.2 Zigzag cavity

4 Hybrid system description and dynamics

References

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